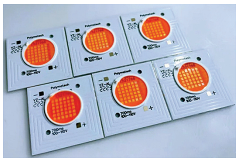
NEW PRODUCTS

MAKE IN INDIA

COMPONENTS

LEDs to improve farming

Polymatech Electronics has released Ravaye, a series of LEDs for smart horticulture. The monochromatic LEDs can enhance greenhouse and

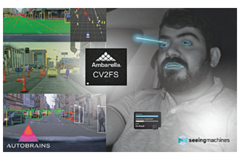


vertical farming lighting for professional horticulture applications. The LEDs produce a range of wavelengths that can promote healthier plant growth by accelerating photosynthesis, strengthening plant immunity, and increasing nutritional value. The Ravaye LEDs offer a wide variety of wavelength combinations to suit different plant needs. Their light can enhance plant growth, improve farming environments, and can also help lower a farmer's lighting system costs, thus increasing profitability for the farmers.

Polymatech Electronics
www.polymatech.in

SoC with ADAS and DMS

This system-on-chip (SoC) combines advanced driver assistance systems (ADAS) with driver and occupant monitoring. The SoC won't only

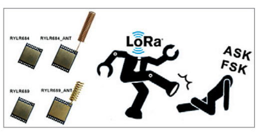


reduce the size of the component but will also lower the cost, complexity, and speed of adding advanced safety systems to a broad range of vehicles. This integrated solution will require low power consumption, reducing thermal-management needs, which is critical for electric vehicles. The low power and high performance capability enable cameras to be packaged in a single box that can be placed in high-temperature areas, thus giving freedom to designers to place the system anywhere in the vehicle. This allows designers to preserve their development investments while tailoring the selected SoC's processing power to the needs of each car model's features, from basic ADAS and driver monitoring system (DMS) to full domain control using the 5nm CV3 family for Level 2+ to L4 autonomy and beyond.

Seeing Machine, https://seeingmachines.com
Autobrain, https://www.autobrains.ai
Ambarella, https://www.ambarella.com

Long-range transceiver modules

The RYLR684 and RYLR689 from Reyax Technology are cost-effective LoRa and (G)FSK (Gaussian frequency-shift keying) transceiver modules that are suitable for long-range wireless applications. The cost-effective



LoRa modules are based on a Semtech LLCC68 chip and have a programmable transmission power level from -20dBm to 22dBm. This allows the designers to optimise the range and power of the transmitter for each use case, thus reducing the energy consumption of battery-powered devices. This replaces the traditional frequency shift keying (FSK) and amplitude shift keying (ASK) that had higher energy consumption. To enhance the sensitivity and achieve better performance, both the modules come with metal covering that protects them against EMI interference, and offer considerably better performance even when surrounded by noise from other devices. The LoRa modules are suitable for a wide range of use cases, including asset tracking, smart agriculture, smart buildings, industrial automation, and various other IoT applications.

Reyax
https://reyax.com

Timekeeper for aerospace and defense

SiT7910 from SiTime Corporation is a 32kHz MEMS Super-TCXO, a timekeeping component used in electronics. The SiT7910 is part of the SiTime Endura ruggedised family for aerospace and defence applications. The Super-TCXO offers up to 25x higher precision, which

enables the devices that use GPS signals such as ruggedised handheld radios, ground vehicles, and sensors to lock to GPS more quickly and securely. The SiT7910 measures just 2.5mm × 2.0mm and works with a voltage from 1.62V to 3.63V. It offers a high 20ppb/g g-sensitivity for resilience in high-vibration environments. Moreover, it has ultra-low power consumption, typically 6.0µA, for use in portable aerospace and defense electronics. The Super-TCXO has been designed for long lasting usage and hence it has a ±5ppm frequency aging over 20 years, so the device can maintain its performance for over a decade.

SiTime Corporation
www.sitime.com

Gate driver IC for EV inverters

The RAJ2930004AGM from Renesas is a gate driver IC for driving EV invert-